Bridging Biology and AI: Vision-Language Models for Protein Function Understanding

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Introduction

Problem: Protein function is hard to determine
- Sequence data grows faster than experimental annotation
- Traditional methods (structure/homology) are slow or infeasible for novel proteins

Opportunity:
- Large language models (e.g., ESM-3) create embeddings that capture functional signals
- Opens door to faster, scalable predictions

This Project:
- Use vision-language AI (BLIP-2) to interpret ESM-3 embeddings as images
- Generate natural language descriptions of protein function
- Compare predictions to UniProt and Gene Ontology

Goal:
- Make protein function prediction faster, interpretable, and accessible
- No 3D models or lab work required

Methodology

Goal:
- Test if visualizing protein sequences improves interpretability

Approach:
- Convert protein sequences → embeddings (ESM models)
- Reshape high-dimensional outputs into 2D matrices
- Render as normalized .png images

Evaluation:
- Compare AI-generated function descriptions with known annotations
- Use sentence-transformer embeddings + cosine similarity for accuracy checks

Figure 3 (a)

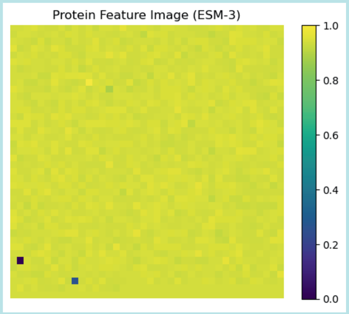


Figure 3 (b)

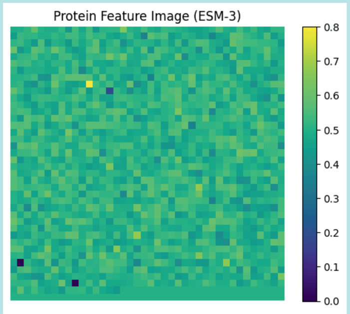


Figure 1 (a)

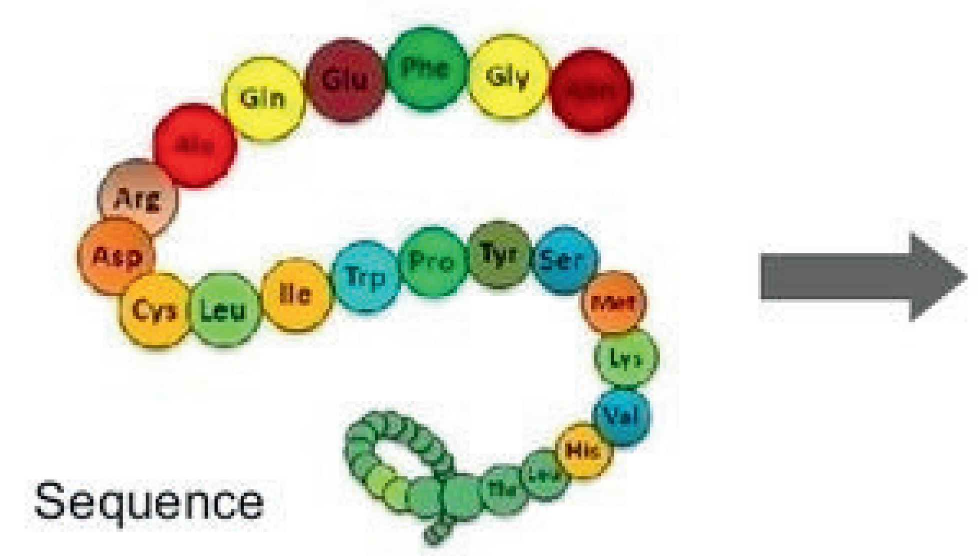


Figure 1 (b)

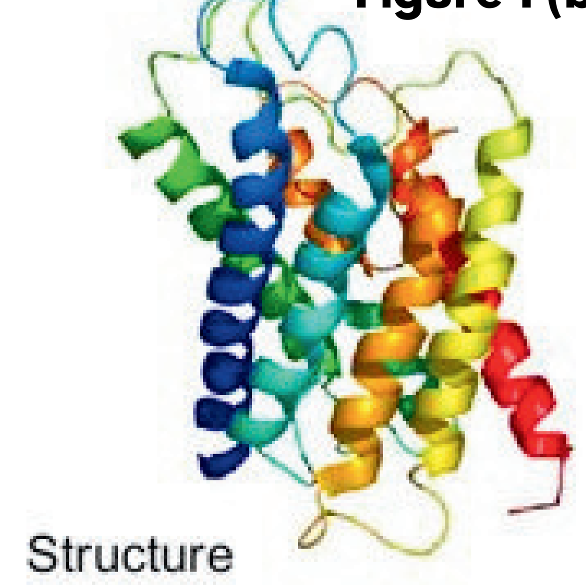


Figure 1 (c) **Figure 1 (d)**

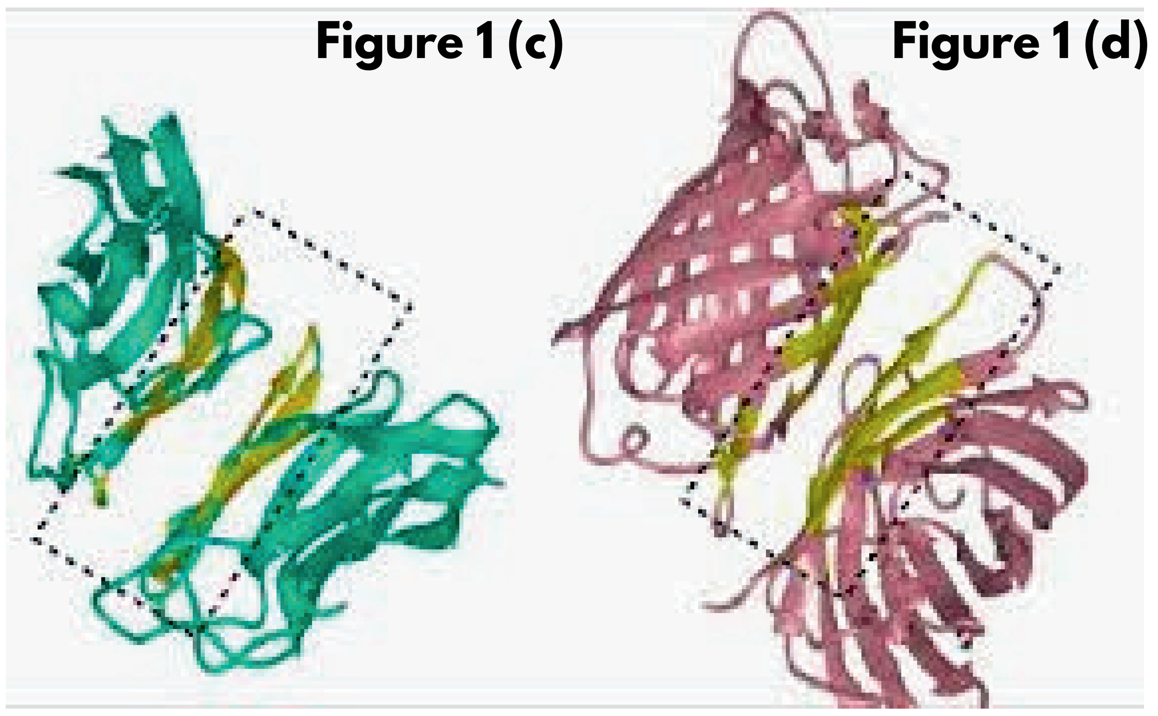
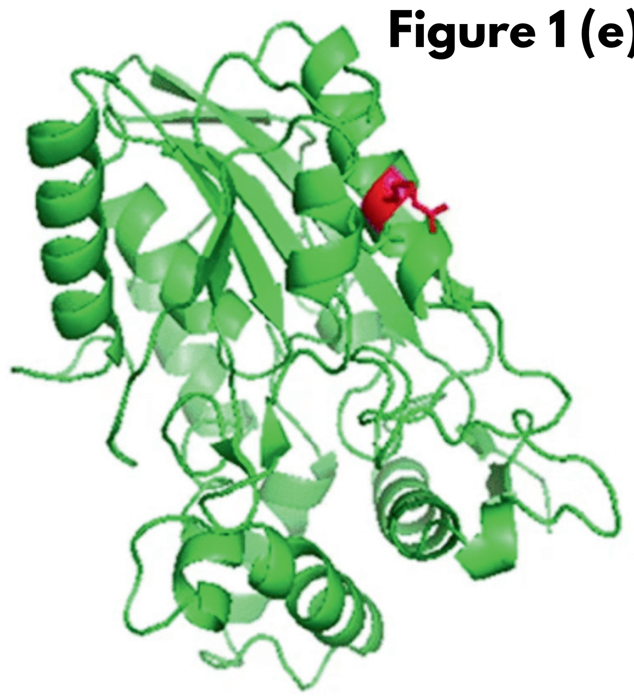


Figure 1 (e)



Flowchart Piplelines

Raw Protein Sequence (FASTA)
↓
[ESM-3 Embeddings]
↓
[2D Feature Matrix]
↓
[Protein Image (.png)]
↓
[Image-to-Text Model: BLIP-2]
↓
[Generated Description] → [Similarity Scoring]
↓
[GO / UniProt Reference Description] → [Similarity Scoring]
↓
[Accuracy Score]

Figure 2 (a)

M K T F FVLLCTFTVSGTANAQDNPKTITISNDGTY

tensor([
[0.023, -0.117, 0.005, ..., 0.021, -0.033, 0.041],
[0.015, -0.098, 0.012, ..., 0.010, -0.021, 0.027],
[0.019, -0.102, -0.007, ..., 0.034, -0.029, 0.020],
[0.017, -0.095, -0.005, ..., 0.038, -0.024, 0.019], ...])

columns: latent feature dimension (out of 512)
values: learned numeric feature describing residue context

Results

Figure 4 (a)

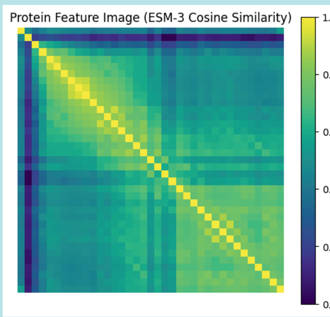


Figure 4 (b)

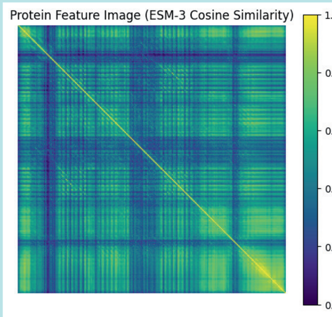


Figure 4 (c)

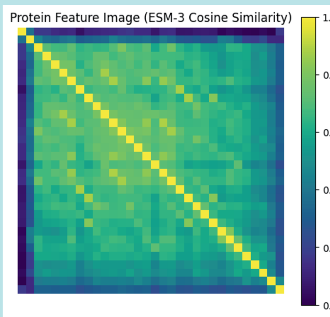
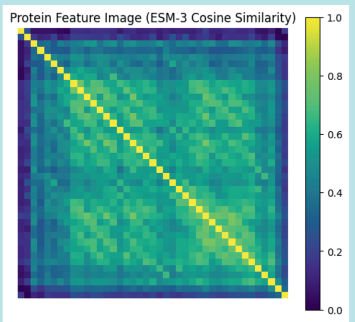


Figure 4 (d)



Generated Text Example: The 2D similarity matrix shows a strong diagonal of residue self-similarity and block-like clusters, indicating multiple structured domains or tandem repeats. Off-diagonal patches suggest inter-domain interactions or conserved motifs. Overall, the protein is modular and likely adopts a globular, multi-domain fold.

Conclusion

Novel Pipeline:
- Translates protein sequences → natural language
- Combines ESM-3 embeddings + BLIP-2 (vision-language AI)
- Uses 2D matrix reshaping + image rendering for interpretation

Current Progress:
- Pipeline complete through image generation stage
- ESM-3 integrated with custom reshaping methods
- Framework ready for next processing stages

Next Steps:
- Fine-tune 2D feature matrices
- Add batch-level processing
- Implement BLIP-2 image-to-text stage
- Evaluate vs. UniProt & Gene Ontology (embedding-based similarity)

Impact:
- Shows technical feasibility of visual protein representation
- Offers a scalable, interpretable alternative to structure-based methods
- Bridges protein language models with multimodal AI

Key Sources & Acknowledgements

ProteinNet – Standard dataset for ML on structures https://arxiv.org/abs/1902.00249
AlphaFold2 Structural Models of Complexes https://www.researchgate.net/figure/AlphaFold2-structural-models-of-protein-complexeswithout-experimentally-determined-3D_fig1_354610833
Deep Learning in Structural Bioinformatics (Bib) https://academic.oup.com/bib/article/22/6/bbab278/6332321
Bioinformatics Visualization Tools (Bib) https://academic.oup.com/bib/article/25/4/bbae349/7717955
Protein Function Prediction Tools (Bib) https://academic.oup.com/bib/article/11/1/153/194484
Seeing the PDB – Visualization Techniques https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8167287/ https://academic.oup.com/bib/article/25/5/bbae359/7720609?searchresult=2

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